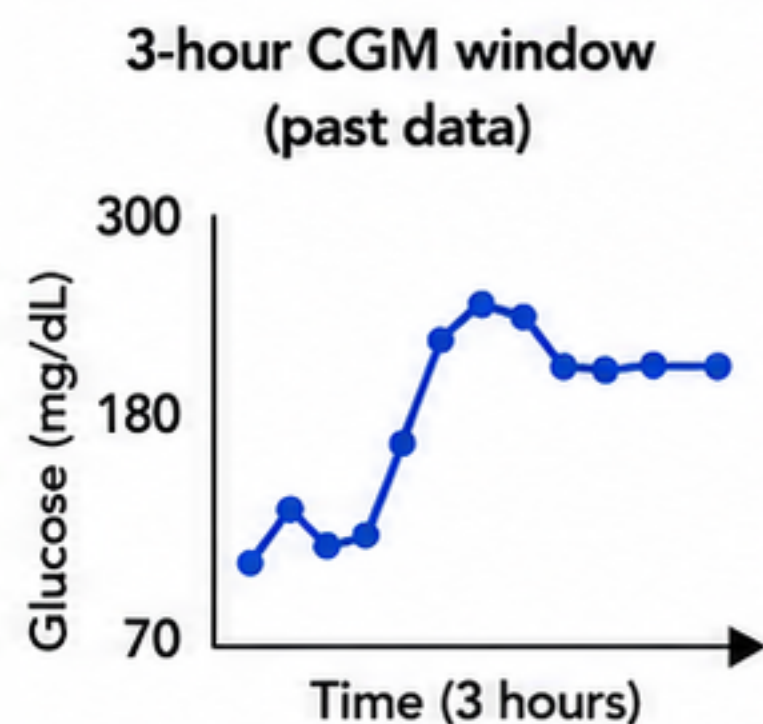


Behavioural Feedback & Decision Support



1 CGM Window Extraction



CGM provides glucose readings every 5 minutes.

2 LLM Agent: Contextualization

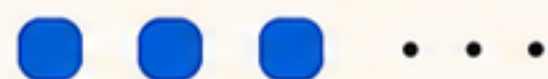


LLM AGENT
(e.g., GPT-4)

Generates morphology-aware textual summary from CGM window.

Example Summary
Gradual **upward trend** interrupted by brief **plateau** and small **dips**.

Text Embedding



3 Multimodal Encoding

Glucose Patches



Patch Embedding



Text Embedding

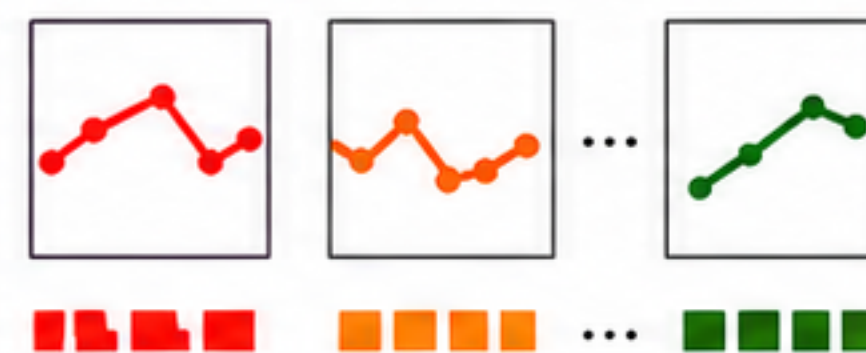


Multimodal Encoder
(Cross-Modal Fusion)

Glucose patches and text embedding are fused in a multimodal transformer.

4 Retrieval-Augmented Module

Episode Library
(Historical Patterns)

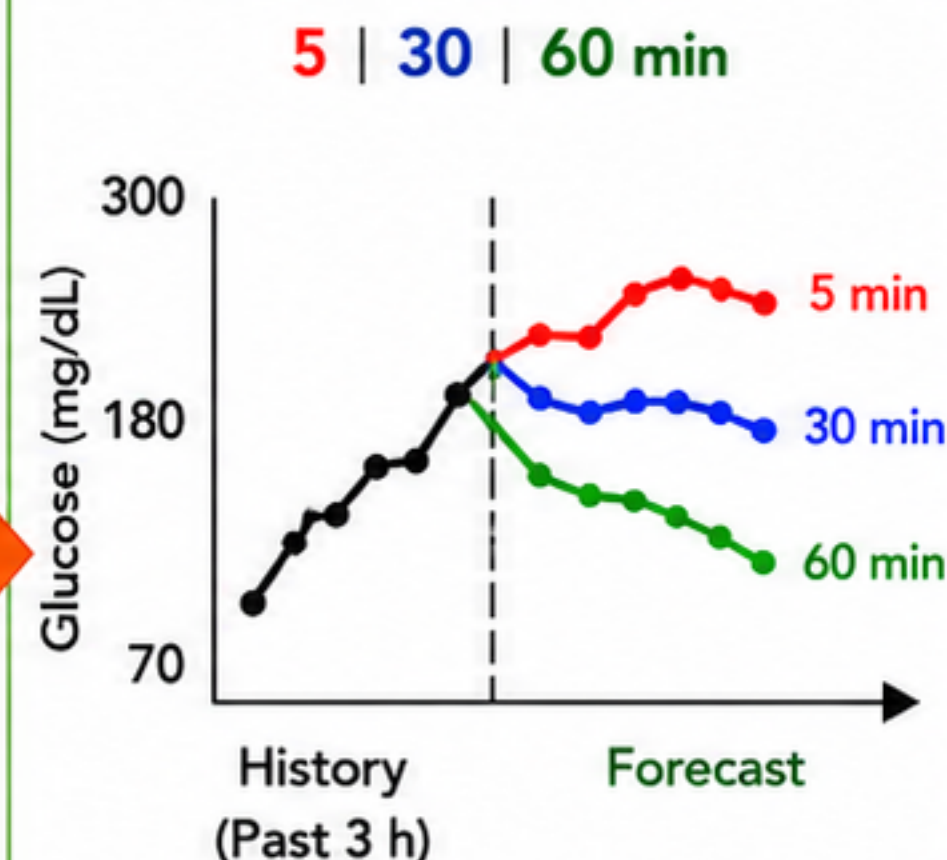


Similarity Search
(Top-K Retrieval)

Cross-Attention
(Fuse Similar Episodes)

Retrieves similar historical patterns and attends to them to provide contextual guidance.

5 Multi-Horizon Forecast



Decision Support

- ✓ Early warning of highs/lows
- ✓ Treatment adjustment
- ✓ Proactive interventions